

Exercises on variable change

Practice for the exam on June 1, 2026

Review: Variable change in higher dimension for the Lebesgue measure

Recall the theorems around change of variables in $\mathbb{R}^d$:

Theorem 1 (Change of variables in $\mathbb{R}^d$). *Let D_1, D_2 be open subsets of $\mathbb{R}^d$, $\phi : D_1 \rightarrow D_2$ a $\mathcal{C}^1$ -diffeomorphism and $f : D_2 \rightarrow \mathbb{R}$ a function. The following assertions are true:*

- a) *f is positive and measurable on D_2 if and only if $(f \circ \phi)|\det J_\phi|$ is positive and measurable on D_1 ;*
- b) *f is integrable on D_2 if and only if $(f \circ \phi)|\det J_\phi|$ is integrable on D_1 .*

where J_ϕ is the Jacobian matrix of ϕ . Moreover, whenever f is positive measurable or integrable, we have:

$$\int_{D_2} f(y) \, dy = \int_{D_1} f(\phi(x)) |\det J_\phi(x)| \, dx$$

This appears cryptic in essence: how do I, as a student, know where to multiply by $|\det J_\phi|$? Or which function ϕ is?

To remember the big idea, it can be good to draw a parallel with single-variable calculus. Typically, if I have the integral $\int f(x) \, dx$ (where $x \in \mathbb{R}^d$) and I wish to apply the change of variables $y = \phi(x) \in \mathbb{R}^d$, then I can take the very simple single-variable $\phi : x \mapsto 2x$ for some function $f : \mathbb{R} \rightarrow \mathbb{R}$. In particular, ϕ is bijective (its reciprocal is $\phi^{-1} : y \mapsto y/2$) and the change of variables $y = \phi(x) = 2x$ gives

$$y = 2x \quad \text{so} \quad \frac{1}{2} \, dy = dx \quad \text{so} \quad \int f(x) \, dx = \int \frac{1}{2} f\left(\frac{y}{2}\right) \, dy$$

Note that here, the coefficient applied is $1/2$, which you can check is equal to $|\det J_{\phi^{-1}}|$. In the same fashion, this example can help to remember:

$$y = \phi(x) \quad \text{so} \quad |\det J_{\phi^{-1}}(y)| \, dy = dx \quad \text{so} \quad \int f(x) \, dx = \int |\det J_{\phi^{-1}}(y)| f(\phi^{-1}(y)) \, dy$$

Note however that **under no circumstance are you to reason this way on your exam paper**. Analogies like this are exclusive to your scrap paper, and only serve to get your ideas straight. Otherwise, you only write the variable change on the exam paper.

To properly execute a change of variables on an exam paper:

- a) Begin by writing out ϕ . Prove that it is bijective, and give its reciprocal if you can express it simply.
- b) Prove that both ϕ and ϕ^{-1} are $\mathcal{C}^1$. You have various methods to do so.
- c) Find out if the variable change you are looking to execute uses ϕ , or ϕ^{-1} . This is when a single-variable analogy can help.

d) Figure out where you are multiplying by $|\det J_\phi|$, and execute the variable change.

Lastly, note that **polar and spherical coordinate changes** require less justification, as you already have access to these results in your coursework. There is no point in writing out the explicit ϕ function or proving that it is a $\mathcal{C}^1$ -diffeomorphism (you will suffer if you attempt to do so).

Exercises

This first exercise focuses on doing a change of variable for single-variable functions. Although not necessary here, it is still good practice to show that ϕ is a $\mathcal{C}^1$ -diffeomorphism.

Question 1. Using two variable changes, show:

$$\int_0^1 \frac{\sin(\log x)}{x \log x} dx = \int_0^{+\infty} \frac{\sin u}{u} du$$

For your knowledge, this integral evaluates to $\pi/2$ (Dirichlet integral).

This next (pseudo-classic) exercise makes you review polar coordinate changes.

Question 2. Using a polar coordinates, after having justified that you may do so, evaluate:

$$I := \int_{-\infty}^{+\infty} e^{-x^2/2} dx = ?$$

HINT - What does I^2 look like (using Fubini)?

Question 3. Let $a \in \mathbb{R}$ and $b \in \mathbb{R}^d$ be fixed, for some integer $d \geq 1$. Define $\phi : x \in \mathbb{R}^d \mapsto a \cdot x + b$ where $a \cdot x$ is scalar multiplication. Show that ϕ is a $\mathcal{C}^1$ -diffeomorphism if and only if $a \neq 0$. What is the determinant of its Jacobian matrix equal to? What if $a = 0$ - is ϕ still differentiable? What is the determinant of its Jacobian matrix equal to then?

Question 4. Using the previous question, show that the one-dimensional Lebesgue measure on $\mathbb{R}$ is translation-invariant, *i.e.*

$$\forall x \in \mathbb{R}, \forall B \in \mathcal{B}(\mathbb{R}), \quad \lambda(x + B) = \lambda(B)$$

where $x + B := \{x + b : b \in B\}$ by definition. Can you use the previous question to find a relation between $\lambda(aB)$, a , and $\lambda(B)$?

Question 5. Using a variable change, show:

$$\int_0^1 x \sqrt{1-x^4} dx = \int_0^{\pi/2} \frac{1}{2} \sin(\theta)^2 d\theta$$

Question 6. Show that the following ϕ is a $\mathcal{C}^1$ -diffeomorphism for the right domain and co-domain.

$$\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}, \quad \phi(x, y) := \left(x^2 + y^2, \frac{x}{\sqrt{x^2 + y^2}} \right)$$

Note that multiple answers can be obtained here (for the domain and co-domain on which ϕ is a $\mathcal{C}^1$ -diffeomorphism).

As the next exercise uses this function again, we will reveal the answer. Both ϕ_1 and ϕ_2 (defined by the above expression) are $\mathcal{C}^1$ -diffeomorphisms.

$$\phi_1 : \mathbb{R} \times (0, +\infty) \rightarrow (0, +\infty) \times (-1, 1)$$

$$\phi_2 : \mathbb{R} \times (-\infty, 0) \rightarrow (0, +\infty) \times (-1, 1)$$

where $(-1, 1)$ is the open segment from -1 to 1 .

Question 7. Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ be a positive and measurable function. Using the previous question, prove:

$$\int_{\mathbb{R}^2} (g \circ \phi)(x, y) e^{-(x^2+y^2)/2} dx dy = \int_{(0,+\infty) \times (-1,1)} g(u, v) \frac{e^{-u}}{\sqrt{1-v^2}} du dv$$

HINT - Recall that $\lambda^{\otimes 2}(\mathbb{R} \times \{0\}) = 0$. You may write $\mathbb{R}^2 = \mathbb{R} \times (0, +\infty) \cup \mathbb{R} \times (-\infty, 0) \cup \mathbb{R} \times \{0\}$.

Question 8. Evaluate, using a change of variables, the integral:

$$\int_0^{+\infty} \frac{\arctan(y)}{1+y^2} dy = ?$$

HINT - You should get $\pi^2/8$.

END of exercises.

Answers

Question 1. Using two variable changes, show:

$$\int_0^1 \frac{\sin(\log x)}{x \log x} dx = \int_0^{+\infty} \frac{\sin u}{u} du$$

For your knowledge, this integral evaluates to $\pi/2$ (Dirichlet integral).

We are going to use the variable change $x = \exp(u)$, writing:

$$\begin{aligned}\phi : x &\mapsto \log x \\ \phi^{-1} : u &\mapsto \exp(u)\end{aligned}$$

with $\phi : (0, 1) \rightarrow (-\infty, 0)$ and $\phi^{-1} : (-\infty, 0) \rightarrow (0, 1)$. $\phi^{-1} = \exp$ is already known to be $\mathcal{C}^1$ over its appropriate domain, and same goes for $\phi = \log$. So the change of variables is justified.

We can then rewrite the first integral by:

$$\int_0^1 \frac{\sin(\log x)}{x \log x} dx = \int_0^1 \frac{\sin(\phi(x))}{x \phi(x)} dx$$

We still need to calculate:

$$|\det J_\phi(x)| = |\phi'(x)| = \frac{1}{x}$$

Thus, applying the variable change immediately gives:

$$\int_0^1 \frac{\sin(\log x)}{x \log x} dx = \int_{(-\infty, 0)} \frac{\sin u}{u} du$$

We then conclude by the linear variable change $u \mapsto -u$, using $\sin(-u) = -\sin(u)$. □

Question 2. Using a polar coordinates, after having justified that you may do so, evaluate:

$$I := \int_{-\infty}^{+\infty} e^{-x^2/2} dx = ?$$

HINT - What does I^2 look like (using Fubini)?

This integral is famously known as the Gaussian integral. Using the hint, let's write:

$$I^2 = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-(x^2+y^2)/2} dx dy$$

The question tells us to use a polar coordinate change, which will allow us to transform $x^2 + y^2$ into r^2 ($x^2 + y^2$ corresponds to $\|(x, y)\|_2^2$ which becomes r^2 in polar coordinates, simply seen on a drawing for example).

The function $f : (x, y) \mapsto \exp[-(x^2 + y^2)/2]$ is positive and measurable, so polar coordinate change is justified (yes, this is enough!). Applying it gives:

$$\begin{aligned}I^2 &= \int_{-\pi}^{\pi} \left(\int_{\mathbb{R}_+} \exp(-r^2/2) r dr \right) d\theta \\ &= \int_{-\pi}^{\pi} -[e^{-r^2/2}]_0^{+\infty} d\theta \quad \text{using the antiderivative} \\ &= \int_{-\pi}^{\pi} 1 d\theta \\ &= 2\pi\end{aligned}$$

We can then conclude by taking the square root on both sides, knowing that $\exp(-x^2/2)$ is a positive function so $I \geq 0$. □

Question 3. Let $a \in \mathbb{R}$ and $b \in \mathbb{R}^d$ be fixed, for some integer $d \geq 1$. Define $\phi : x \in \mathbb{R}^d \mapsto a \cdot x + b$ where $a \cdot x$ is scalar multiplication. Show that ϕ is a $\mathcal{C}^1$ -diffeomorphism if and only if $a \neq 0$. What is the determinant of its Jacobian matrix equal to? What if $a = 0$ – is ϕ still differentiable? What is the determinant of its Jacobian matrix equal to then?

If $a = 0$, then $\phi : x \mapsto b$ is constant – it is differentiable everywhere, with $d\phi = 0$. The determinant of the Jacobian matrix is then equal to 0, and notably ϕ is NOT a $\mathcal{C}^1$ -diffeomorphism.

Suppose now that $a \neq 0$. For $x, h \in \mathbb{R}^d$, we have:

$$\phi(x + h) - \phi(x) = a \cdot h$$

so ϕ is differentiable in x , and:

$$d\phi(x) : h \mapsto a \cdot h$$

whose Jacobian matrix is simply equal to the diagonal matrix with only a on the diagonal.

Let's show that ϕ is bijective by exhibiting a reciprocal function:

$$\phi^{-1} : y \mapsto \frac{1}{a} \cdot (y - b)$$

For the same reasons ($1/a \neq 0$) ϕ^{-1} is also $\mathcal{C}^1$ so it is a $\mathcal{C}^1$ -diffeomorphism. Moreover,

$$|\det J_\phi| = a^d$$

□

Question 4. Using the previous question, show that the one-dimensional Lebesgue measure on $\mathbb{R}$ is translation-invariant, *i.e.*

$$\forall x \in \mathbb{R}, \forall B \in \mathcal{B}(\mathbb{R}), \quad \lambda(x + B) = \lambda(B)$$

where $x + B := \{x + b : b \in B\}$ by definition. Can you use the previous question to find a relation between $\lambda(aB)$, a , and $\lambda(B)$?

Recall previous exercises where we used the identity:

$$\lambda(B) = \int_{\mathbb{R}} \mathbb{1}_B(y) \, dy$$

Consider some fixed $x \in \mathbb{R}$ (we are working with the one-dimensional Lebesgue measure, though this also works in higher dimensions). Writing $\phi : y \mapsto y + x$ (which is a $\mathcal{C}^1$ -diffeomorphism by the previous question applied to $a = 1$ and $b = x$) we have:

$$\begin{aligned} \lambda(x + B) &= \int_{\mathbb{R}} \mathbb{1}_{x+B}(y) \, dy \\ &= \int_{\mathbb{R}} \mathbb{1}_{x+B}(\phi^{-1}(x + y)) \, dy \\ &= \int_{\mathbb{R}} \mathbb{1}_{x+B}(x + y) \, dy \\ &= \int_{\mathbb{R}} \mathbb{1}_B(y) \, dy \\ &= \lambda(B) \end{aligned}$$

From whence the result of translation invariance. Again, this exact method works in any dimension $d \geq 1$.

For $\lambda(aB)$, suppose first that $a \neq 0$. Then $\phi : x \mapsto a \cdot x$ is a $\mathcal{C}^1$ -diffeomorphism by the previous exercise. Through variable change:

$$\begin{aligned} \lambda(aB) &= \int_{\mathbb{R}} \mathbb{1}_{aB}(x) \, dx \\ &= \int_{\mathbb{R}} \mathbb{1}_{aB}(\phi^{-1}(ax)) \, dx \\ &= \int_{\mathbb{R}} \mathbb{1}_{aB}(ax) \, a \, dx \\ &= a \lambda(B) \end{aligned}$$

using the fact that $|\det J_{\phi(x)}| = a$ according to question 3. In higher dimension, what does this identity become, using the same reasoning? $\square$

Question 5. Using a variable change, show:

$$\int_0^1 x\sqrt{1-x^4} \, dx = \int_0^{\pi/2} \frac{1}{2} \sin(\theta)^2 \, d\theta$$

The variable change here is $x = \sqrt{\cos \theta}$. We know that:

$$\phi : \theta \in (0, \pi/2) \mapsto \sqrt{\cos \theta} \in (0, 1)$$

is a $\mathcal{C}^1$ -diffeomorphism (this will not be shown here, but seeing that $\cos$ is a usual function and $\sqrt{\cdot}$ too, no justification is really needed to prove this). Writing:

$$x = \sqrt{\cos \theta} \quad \text{gives} \quad dx = \frac{-\sin \theta}{2\sqrt{\cos \theta}} \, d\theta$$

So:

$$\int_0^1 x\sqrt{1-x^4} \, dx = \int_{\pi/2}^0 -\frac{\sin \theta}{2\sqrt{\cos \theta}} \times \sqrt{\cos \theta} \times \sqrt{1-(\cos \theta)^2} \, d\theta$$

The result is obtained immediately after simplification. As an additional exercise in integration, can you calculate the result of this integral? $\square$

Question 6. Show that the following ϕ is a $\mathcal{C}^1$ -diffeomorphism for the right domain and co-domain.

$$\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}, \quad \phi(x, y) := \left(x^2 + y^2, \frac{x}{\sqrt{x^2 + y^2}} \right)$$

Note that multiple answers can be obtained here (for the domain and co-domain on which ϕ is a $\mathcal{C}^1$ -diffeomorphism).

There is no problem for the left part of ϕ , defined by $(x, y) \mapsto x^2 + y^2$. This can take any (strictly) positive value, meaning that the left part can reach $(0, +\infty)$.

The right part has a singularity in the point $(0, 0)$ which causes some issues for differentiability. Indeed, ϕ isn't even continuous in $(0, 0)$.

Let (x, y) be a point in which ϕ is differentiable. Then we can calculate the Jacobian matrix of ϕ in (x, y) :

$$J_{\phi}(x, y) = \begin{bmatrix} 2x & 2y \\ \frac{y^2}{(x^2+y^2)^{3/2}} & -\frac{xy}{(x^2+y^2)^{3/2}} \end{bmatrix}$$

which gives

$$|\det J_{\phi}(x, y)| = \frac{2|y|}{(x^2+y^2)^{1/2}} = 2\sqrt{1 - \left(\frac{x}{\sqrt{x^2+y^2}}\right)^2} = 2\sqrt{1-v^2}$$

where $\phi(x, y) = (u, v)$.

Note that:

$$\left| \frac{x}{\sqrt{x^2+y^2}} \right| \leq 1$$

So, v takes its values in $(-1, 1)$. In addition, parity relative to y (i.e. $v(y) = v(-y)$) makes it so that any attained value can be attained twice. This evidently conflicts with bijectivity, forcing us to restrain the domain and co-domain of ϕ . Consider:

$$\phi_1 : \mathbb{R} \times (0, +\infty) \rightarrow (0, +\infty) \times (-1, 1)$$

The formal proof that ϕ_1 is bijective is left to the reader. Given that ϕ_1 is polynomial, ϕ_1^{-1} can be expressed as a fraction of polynomial functions with no poles in its domain, making ϕ_1^{-1} of class $\mathcal{C}^1$. It follows that ϕ_1 is a $\mathcal{C}^1$ -diffeomorphism.

The same reasoning applies to ϕ_2 . □

Question 7. Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ be a positive and measurable function. Using the previous question, prove:

$$\int_{\mathbb{R}^2} (g \circ \phi)(x, y) e^{-(x^2+y^2)/2} dx dy = \int_{(0,+\infty) \times (-1,1)} g(u, v) \frac{e^{-u}}{\sqrt{1-v^2}} du dv$$

HINT - Recall that $\lambda^{\otimes 2}(\mathbb{R} \times \{0\}) = 0$. You may write $\mathbb{R}^2 = \mathbb{R} \times (0, +\infty) \cup \mathbb{R} \times (-\infty, 0) \cup \mathbb{R} \times \{0\}$.

We are going to apply a change of variables. The previous question did all of the work, so:

$$\int_{\mathbb{R} \times (0, +\infty)} (g \circ \phi)(x, y) e^{-(x^2+y^2)/2} dx dy = \int_{(0,+\infty) \times (-1,1)} g(u, v) \frac{e^{-u}}{2\sqrt{1-v^2}} du dv$$

We obtain a similar result over $\mathbb{R} \times (-\infty, 0)$. Summing these two makes the $1/2$ factor disappear. Moreover, seeing that $\lambda^{\otimes 2}(\mathbb{R} \times \{0\}) = 0$, the remaining integral is equal to 0. The result follows immediately. □

Question 8. Evaluate, using a change of variables, the integral:

$$\int_0^{+\infty} \frac{\arctan(y)}{1+y^2} dy = ?$$

HINT - You should get $\pi^2/8$.

The change of variables is $y = \tan x$, with $x \in (0, \pi/2)$. $\phi := \tan$ is known to be $\mathcal{C}^1$, and its reciprocal $y \in (0, +\infty) \mapsto \arctan(y)$ is also $\mathcal{C}^1$. Thus, we are working with a $\mathcal{C}^1$ -diffeomorphism, and:

$$|\det J_\phi(x)| = |1 + \tan^2(x)| = 1 + \tan^2(x) = 1 + y^2$$

Thus,

$$\int_0^{+\infty} \frac{\arctan y}{1+y^2} dy = \int_0^{+\infty} \frac{\phi^{-1}(y)}{|\det J_\phi(y)|} dy = \int_0^{\pi/2} x dx = \frac{\pi^2}{8}$$

□

Best of luck on the exam to all of you.

END of answers.